SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CONCEPT MAPPING

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3, BL4)	Assessment:	Assessment Tool 3 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 100 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

100 Marks

Code of Conduct

I ________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25 %	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25 %	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20 %	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20 %	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10 %	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Q3						
Q4						
Q5						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department

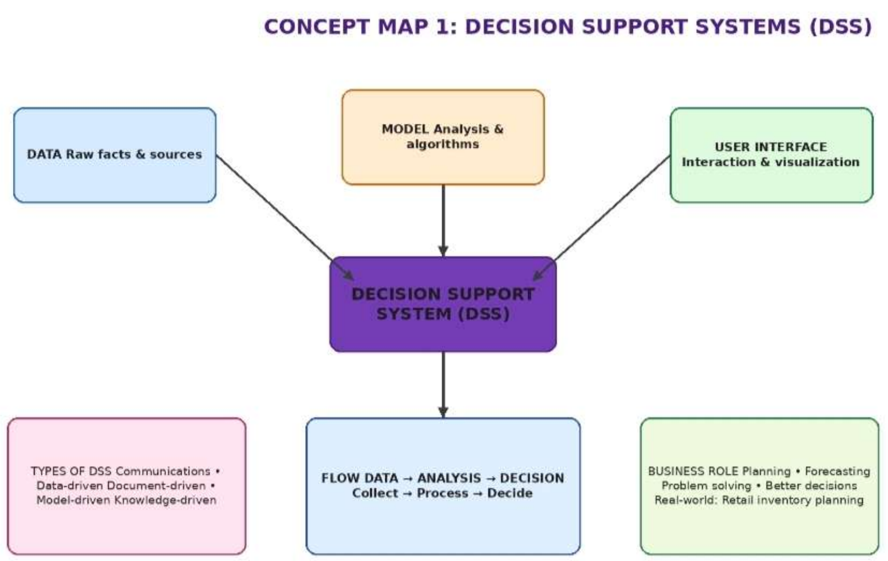
Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Q3						
Q4						
Q5						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

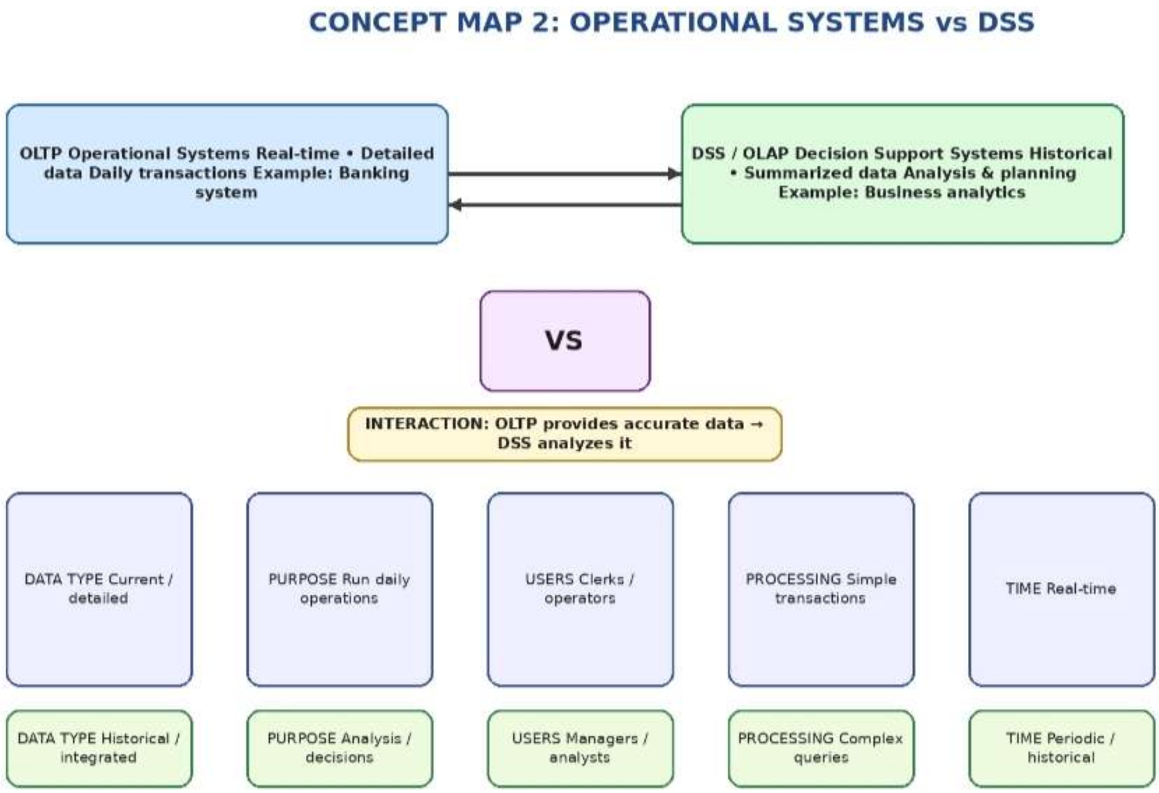
Q1. Decision Support Systems (DSS)

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.



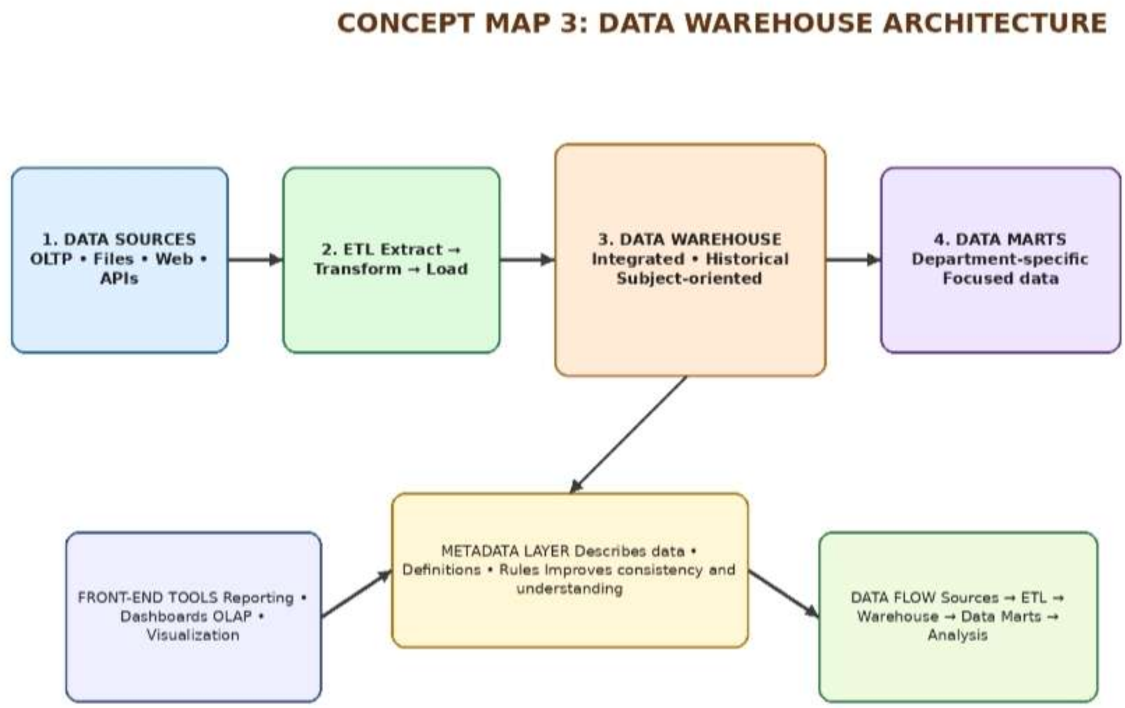
Q2. Operational Systems (OLTP) vs DSS

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).



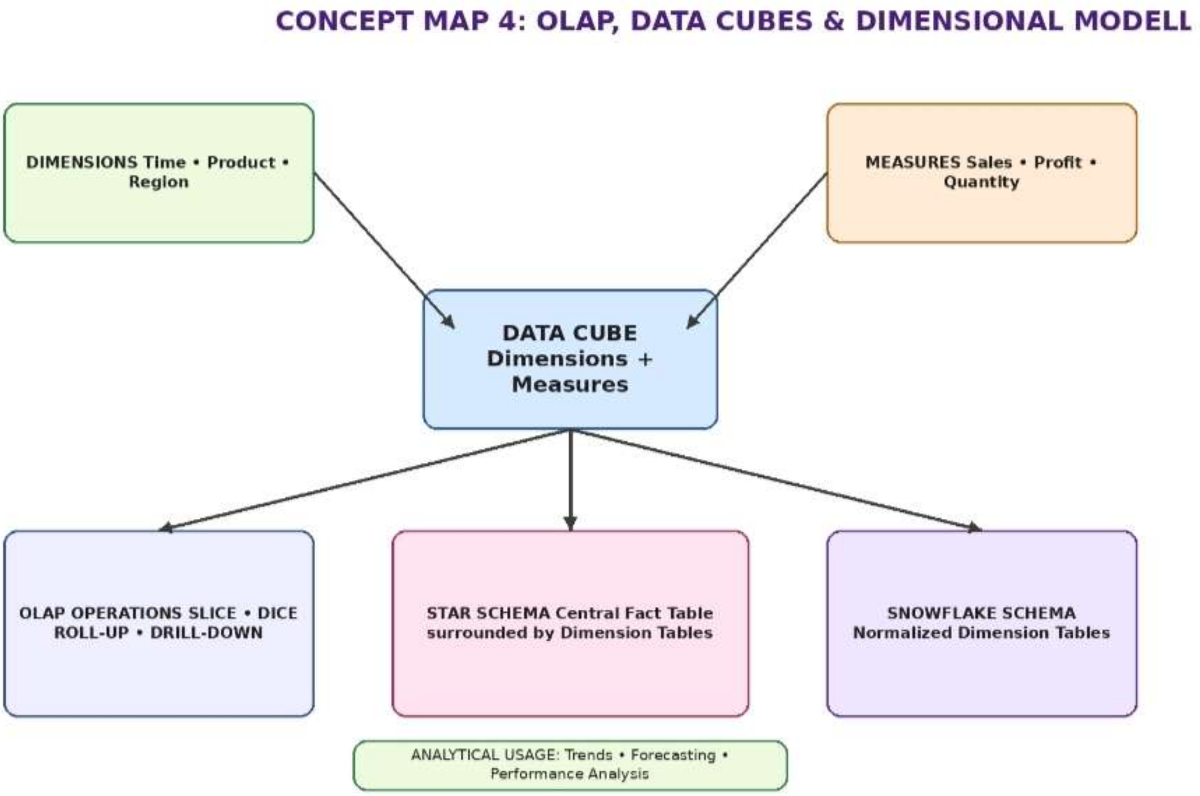
Q3. Architecture of Data Warehouse

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.



Q4. OLAP, Data Cubes & Dimensional Modelling

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modelling techniques.



Q5. Query Processing & BI Tools (KNIME)

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.